INCH-POUND

MIL-PRF-1/1542B(USAF)
18 June 1999
SUPERSEDING
MIL-E-1/1542A(USAF)
21 March 1972

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 7402

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: High vacuum clipper diode or rectifier.

Outline: See figure 1.

Mounting: Vertical position, base down.

Weight: 10 ounces nominal.

ABSOLUTE RATINGS:

Parameter: Unit:	Ef V ac	epx kv	ib a	Ib mA dc	Ieff A ac	Pp W
Clipper						
Maximum:	12.1	33	50	---	1.25	450
Minimum:	10.9	---	---	---	---	---
Rectifier						
Maximum:	12.1	33	2.7	860	---	450
Minimum:	10.9	---	---	---	---	---
Test condition:	11.5	---	---	---	---	---

General:

Qualification - Not required.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:56Z
Check the source to verify that this is the current version before use.

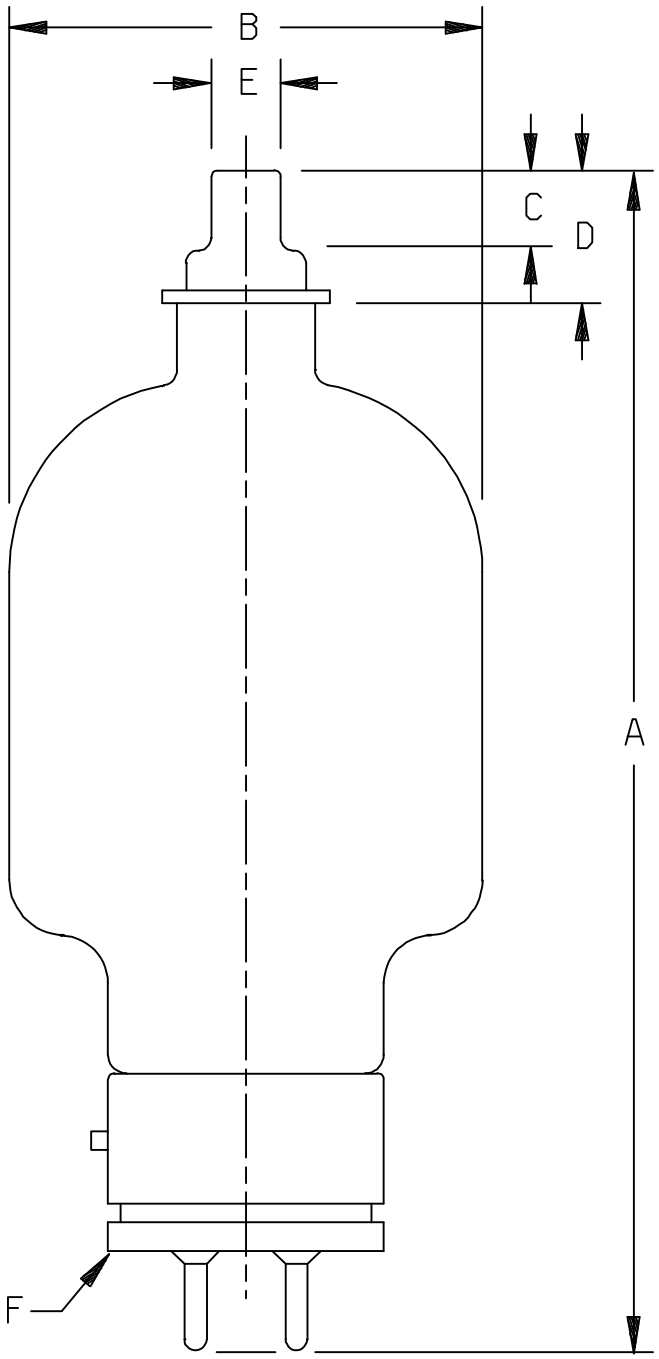
TABLE I. Testing and inspection.

Inspection	Method	Condition	Acceptance Level	Inspection Level	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 1								
Filament current	1301		0.65	II	If	14.75	16.25	A ac
Peak emission	1231	eb = 8.0 kv	0.65	II	is	30	---	a
Electrode current (anode)	1256	Eb = 330 V dc	0.65	II	Ib	860	---	mA dc
Operation of rectifiers	1353	epx = 33 kv; ib = 2.7 a Ib = 860 mA dc 1/	0.65	II	---	---	---	---
Conformance inspection, part 2								
Low-frequency vibration	1031	No voltages	6.5	S3	---	---	---	---
Shock	1041	Hammer angle 25°	---	---	---	---	---	---
Operation	---	epx = 33 kv tp = 1 µsec; Zo = 53 ohms; Rs = 300 ohms (min); Prr = 340 R1 = 0 1/ 2/	---	---	ib	50	---	a
Conformance inspection, part 3								
Life test provisions	---	Group D; operation of rectifiers	---	---	t	500	---	hr
Life-test end points:	---							
Peak emission	1231		---	---	is	25	---	a

1/ The tube shall operate for 2 minutes without arcing or signs of gas during a test interval not to exceed 5 minutes.

2/ The tube shall be operated as a clipper diode in a thyratron modulator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:56Z
Check the source to verify that this is the current version before use.



Dimensions				
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	9.250	9.750	235.15	247.65
B	---	3.625	---	92.07
C	.594	.656	15.09	16.66
Conformance inspection, part 3 (periodic check)				
E	.559	.573	14.20	14.60
F	Base A-4-18			
Reference dimensions				
D	1.00		25.40	

FIGURE 1. Outline drawing of tube type 7402.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:56Z
Check the source to verify that this is the current version before use.

Custodians:
Air Force - 11
DLA - CC

Review activities:
Air Force - 17, 80, 99

Preparing activity:
DLA - CC

(Project 5960-3523)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:56Z
Check the source to verify that this is the current version before use.